Name: ______________________

Class: ______________________

Date: ______________________

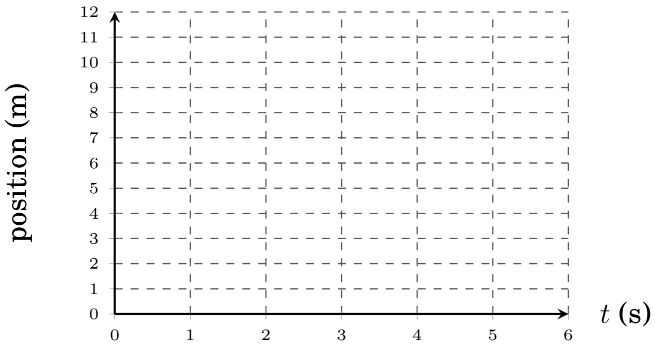
Mr. Rawson • GCSE Physics

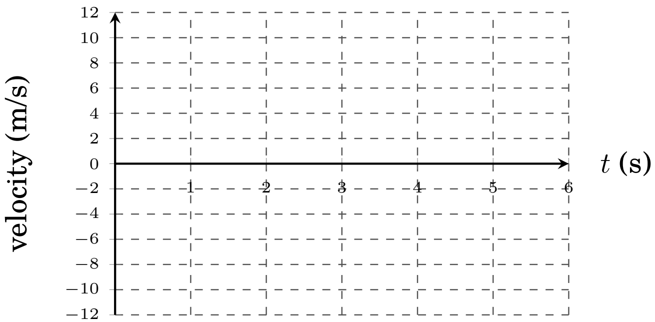
Kinematics Graphing

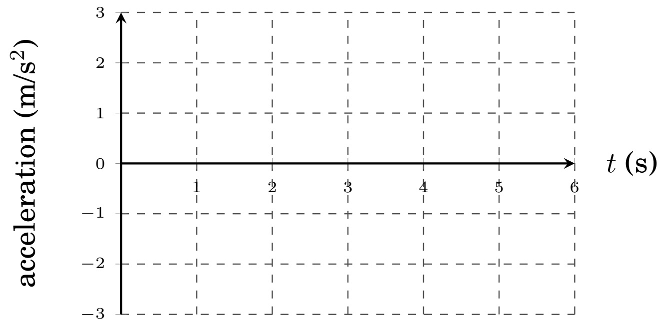
The Basic 6. On the board is a moped that moves in six different ways. For each one, sketch the **three** graphs that describe it — position–time, velocity–time, acceleration–time. The axes are the *same* ones on screen, so copy what you see, and **write down the three numbers at the top of the screen first** — every line depends on them. *Straight, curved, or flat?* is the whole question; for velocity and acceleration, also ask whether the line sits **above or below** the time axis.
The Physics Classroom — Kinematic Graphing

1. Move with Constant Speed in + Direction

Initial velocity ________ m/s • Acceleration ________ m/s^2 • Maximum time ________ s

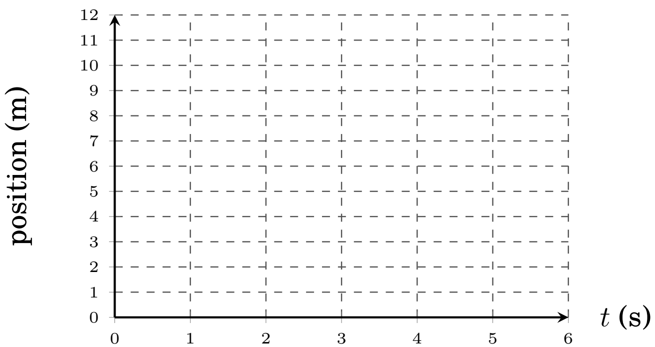


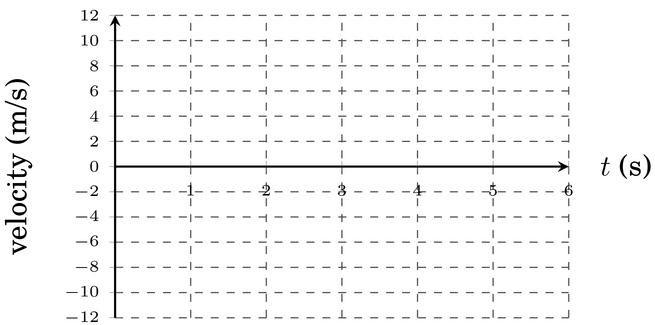


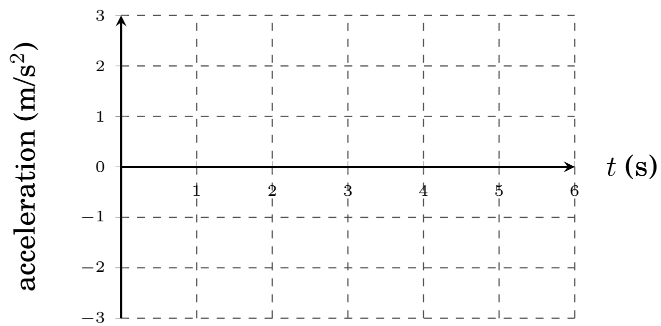


2. Move with Constant Speed in – Direction

Initial velocity ________ m/s • Acceleration ________ m/s^2 • Maximum time ________ s

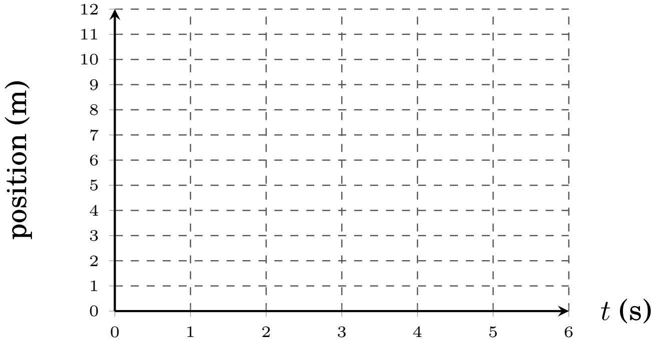


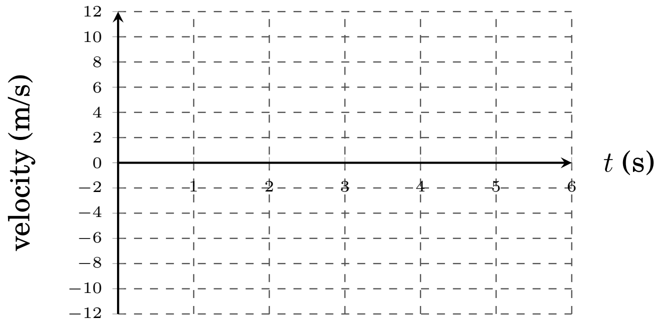


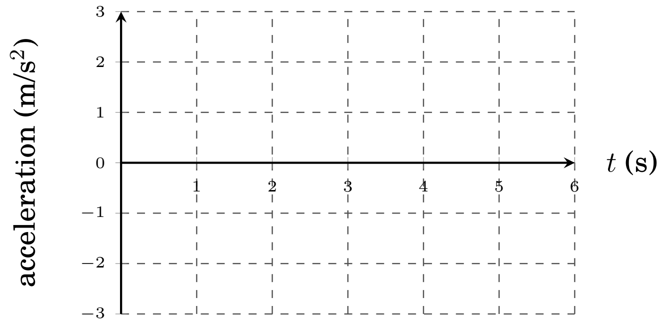


3. Move in + Direction; Speeding Up

Initial velocity _______ m/s • Acceleration _______ m/s^2 • Maximum time _______ s

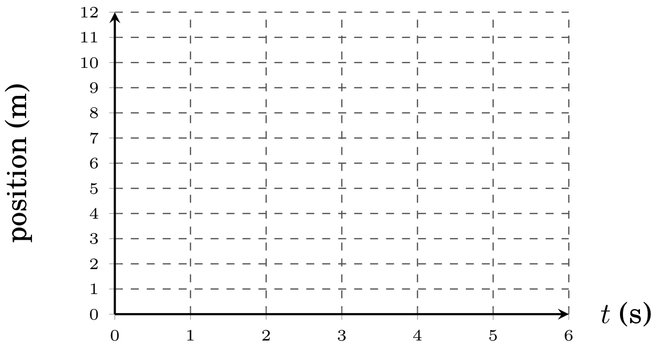


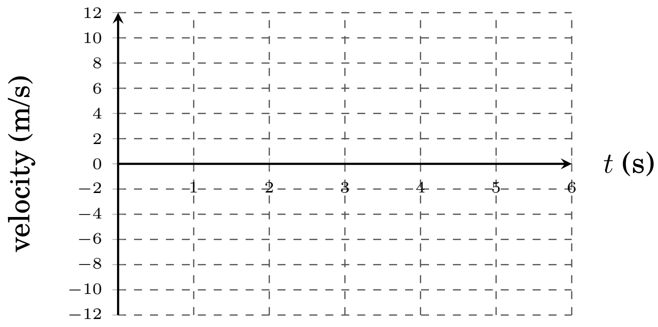


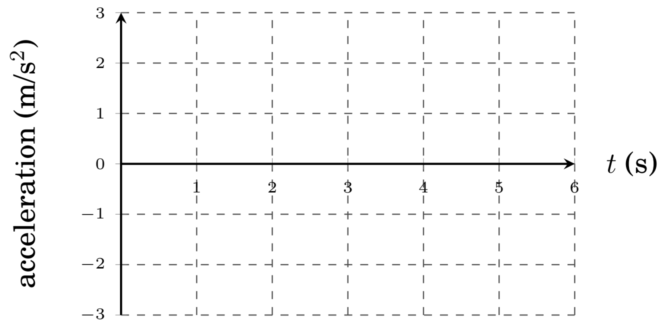


4. Move in + Direction; Slowing Down

Initial velocity _______ m/s • Acceleration _______ m/s^2 • Maximum time _______ s

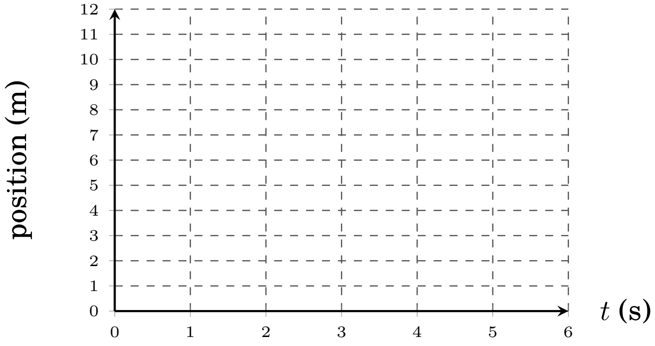


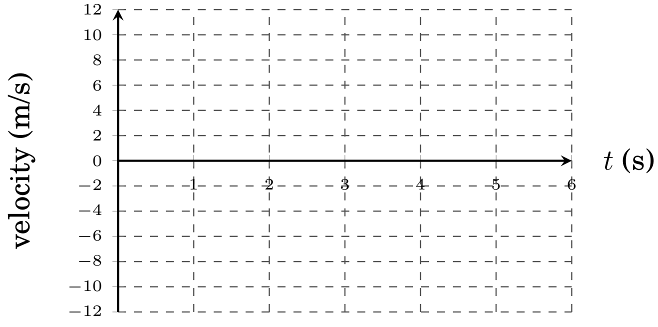


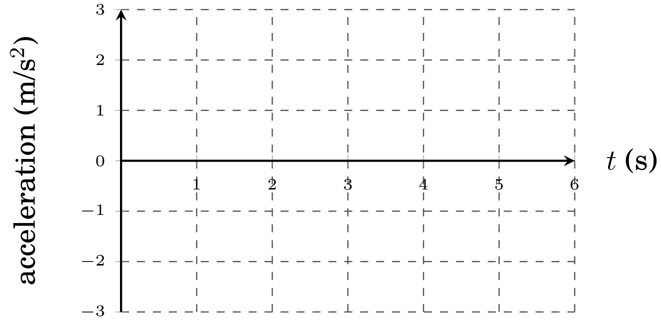


5. Move in – Direction; Speeding Up

Initial velocity ________ m/s • Acceleration ________ m/s^2 • Maximum time ________ s







6. Move in – Direction; Slowing Down

Initial velocity ________ m/s • Acceleration ________ m/s^2 • Maximum time ________ s